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A Novel Model and Decomposition Approach for the Integrated Airline Fleet Assignment, Aircraft Routing, and Crew Pairing Problem

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Given a daily flight schedule and a set of aircraft fleets, the airline scheduling problem assigns individual aircraft and groups of crew to each flight based on specific considerations of aircraft maintenance requirements and crew work rules, respectively. Traditionally, this problem has been sequentially broken down into several stages, where the fleet assignment problem, which is solved first, partitions the entire flight network into subnetworks according to fleet types, followed by respectively solving the aircraft routing and crew pairing problems to generate suitable aircraft and crew rotations. However, this sequential approach ignores the interdependencies among the stages, leading to suboptimal, or even infeasible, solutions. In this paper, we propose an integrated model and solution approach that incorporates the fleet assignment (with itinerary-based demands), aircraft routing, and crew pairing problems within a single framework. We solve the resulting formulation of the problem by using a Benders decomposition approach, along with several acceleration strategies. Computational results obtained by using real-life data from a major U.S. airline demonstrate the benefits of the integrated approach.

Keywords: integrated airline scheduling problem; mixed-integer linear program; Benders decomposition; deflected subgradient optimization; reformulation-linearization technique (RLT); branch-and-price
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1. Introduction

The airline scheduling problem addresses maximizing the total profits or minimizing the total operational costs associated with a flight schedule while satisfying customer demands under a series of restrictions on aircraft maintenance and labor work rules. Specifically, the airline scheduling problem can be broken down into four stages: *schedule planning* (i.e., designing flight schedules), *fleet assignment* (i.e., assigning aircraft fleets to flight legs), *aircraft routing* (i.e., generating maintenance-feasible rotations for aircraft of each type), and *crew pairing* (i.e., generating suitable flight sequences for crews to serve), wherein the output of one stage is sequentially fed into the succeeding stage to achieve implementation.

Once a flight schedule is determined, fleets of different types of aircraft are assigned to different flight legs to maximize the fare-based revenue minus the burned fuel cost and the cost for *spilled* (or lost) demand depending on the assigned fleet's capacity. Here, an *aircraft type* refers to a certain model of aircraft. All aircraft of the same type are identical, i.e., they have the same cockpit configuration, crew rating, and capacity. Furthermore, an *aircraft family* is a set of aircraft types having the same cockpit configuration and crew rating, but different seating capacities. The outcome of the fleet assignment stage partitions the flight network into subnetworks corresponding to aircraft types or families, so that the two subsequent stages of aircraft routing and crew pairing are solved for each specific type or family of aircraft, respectively.

The aircraft routing problem, which is solved next, finds cyclic *maintenance feasible routes* for the aircraft in the fleet while satisfying mandatory maintenance checks, with the objective of optimizing a selective combination of *through-values* (i.e., additional revenue accruing from the same aircraft serving consecutive flight legs), *short connection penalties* (i.e., additional cost due to insufficient intermediate rest or *sit-time* for crews), and/or maintenance costs. Note that, in many cases, the aircraft routing problem is treated as a pure feasibility problem. In addition, a maintenance feasible aircraft route (or *rotation*) is composed of a sequence of flight legs served by a single aircraft which satisfy the following restrictions:

- the total flying time between maintenance checks should not exceed a specified limit;

• the total number of takeoffs between maintenance checks should not exceed a specified maximum value;

• the number of days elapsed between maintenance checks must not exceed a given number of calendar days.

The crew pairing stage requires building a minimum-cost set of pairings such that each leg is covered at least once. Because cockpit crews are qualified to serve only one aircraft family, the crew pairing problem (CP) is solved separately for each aircraft family. In this context, a *duty period* refers to a single workday of a crew serving a sequence of flight legs with short rest periods, or *sits*, separating them, including briefing and debriefing periods at the beginning and end of the period. Crews are assigned to a sequence of flight legs subject to several work-rule requirements established by the *Federal Aviation Administration* (FAA) and union contracts (see Klabjan 2005). For instance, a nonexhaustive list of rules includes the following restrictions, where typical values are specified for the sake of illustration:

• There should be no more than four flight legs in a duty;

• The *minimum sit-time* should be at least 45 minutes. However, if both flight legs are covered by the *same* aircraft, then the sit-time could be reduced to the predefined minimum short connection duration, which is typically 30 minutes (note that this requirement relates aircraft routing decisions to crew scheduling decisions);

• The maximum sit-time in a duty should not exceed four hours;

• The total flying time within a duty period must not exceed eight hours;

• The number of duty periods in a pairing must be at least one and at most four;

• The overnight rest time must be at least 10 hours;

• The total *time away from base* (TAFB) should not exceed 96 hours.

The cost of a duty period, expressed in hours, is the maximum of three quantities, i.e., a certain fraction of the duty period duration, the total flying time, and a specified minimum number of hours. Similarly, the cost of a pairing, expressed in hours, is the maximum of three quantities, i.e., a certain fraction of the pairing duration, the sum of the costs of the duty periods comprising the pairing, and the specified minimum number of hours per duty period times the number of duty periods in the pairing.

Although the foregoing operational phases are sequentially implemented in practice, their interdependence would naturally lead a purely sequential decision-making approach to suboptimal solutions; prefixed decisions made while ignoring downstream considerations would tend to suboptimally restrict the ensuing decision stages. This might even result in infeasibility at some subsequent stages in the process. From this perspective, it is prudent to investigate models that integrate the different stages (or suitable combinations thereof) within a single framework to obtain improved solutions, while being cognizant of the fact that the problem complexity will also substantially increase as more aspects and decisions are simultaneously considered.

In this paper, we propose a novel modeling and algorithmic approach for solving the integrated airline scheduling model that simultaneously considers the operational stages of fleet assignment, aircraft routing, and crew pairing. Note that Papadakos (2009) addressed a similar problem (see §2 for details) wherein set partitioning based formulations were adopted for aircraft fleeting and routing as well as for crew pairing. Consequently, the solution algorithm involves an extra level of the branch-and-price routine within the framework of the master program. However, our model explicitly incorporates maintenance routing requirements, thus resulting in a polynomially-sized formulation for the master program. This significantly simplifies the solution process. Additionally, our model includes itinerary-based demands to improve the quality of fleeting decisions. Moreover, we adopt an alternative effective technique for generating *Pareto-optimal* cuts to accelerate the convergence of the Benders decomposition algorithm.

Specifically, our work makes the following contributions:

1. We propose a novel formulation for the integrated airline scheduling model that incorporates aircraft fleeting, routing, and crew pairing operations in a single framework, in which the fleeting and routing decisions are modeled using a polynomially-sized node-arc flow network representation, and crew pairing decisions are modeled using the traditional set partitioning approach. In addition, we incorporate in the model several realistic operational considerations, such as itinerary-based demands and various mandated aircraft maintenance requirements and crew work rules.

2. We design an effective solution strategy for the developed large-scale model using a Benders decomposition approach that incorporates nondominated Benders cuts along with several acceleration techniques to enhance solvability. Moreover, for the crew pairing subproblem, we adopt a perturbed Lagrangian dual approach along with a specialized deflected subgradient optimization scheme (Subramanian and Sherali 2008) for stabilizing the column generation process. We embed this within a branch-and-price framework.

3. We provide extensive computational results using realistic data obtained from a U.S.-based legacy airline carrier to demonstrate the efficacy of our modeling approach and solution methodology, and we exhibit the benefits of adopting an integrated approach as

opposed to a sequential decision process. The results show that this yields an average of 8.4% improvement in profits.

The remainder of this paper is organized as follows. In §2, we review the literature on existing integrated models for airline operational problems. We present our proposed integrated formulation in §3. We develop several model enhancement techniques in §4, and design a suitable solution strategy. Computational results using realistic simulated data are presented and discussed in §5. We close the paper in §6 with summarizing remarks and possible future research directions.

2. Literature Review

Whereas numerous papers have addressed enhanced airline fleeting models that accommodate aircraft routing or crew pairing considerations, to our knowledge only two papers use models that effectively capture all three features, i.e., aircraft fleeting, aircraft routing, and crew pairing. The first of these is by Sandhu and Klabjan (2007). They present a model that integrates the fleeting and crew pairing decisions, while representing aircraft maintenance considerations in only an aggregate sense via certain aircraft-count requirements. This model is based on the traditional *time-space network* wherein each node represents an activity or event at a station (flight arrival or departure) and each arc represents a flight leg. The authors argued that since the aircraft maintenance routing process is relatively less complicated (and this is true for hub-and-spoke networks), one can neglect detailed routing decisions within the integrated model and simply include some suitable aggregated aircraft-count restrictions. To this end, they designed a novel set of constraints to link the fleet assignment and crew pairing decision variables that would promote feasible aircraft routings. Moreover, Sandhu and Klabjan (2007) proposed two solution schemes based respectively on using Lagrangian relaxation with column generation and using Benders decomposition to effectively solve the formulated problem. The proposed methods were tested on four instances with sizes ranging from two fleet families and 205 legs to four fleet families and 942 legs. The authors found that the Lagrangian relaxation approach was more robust in terms of objective value improvements than the Benders decomposition methodology, while the computational times for the former method were generally shorter than those for the latter. However, as demonstrated by Papadakos (2009), feasible aircraft rotations are not always guaranteed for the resulting solution.

Another paper that integrates the aforementioned three operational stages is by Papadakos (2009). He proposed a model that minimizes the total cost of fleet assignment, aircraft maintenance routing, and crew pairing, less the revenues from through-flights. Similar to the model of Sandhu and Klabjan (2007), this formulation is also based on a flight time-space network, but adopts a path-based representation that facilitates partitioning the network into separate components for each tail number and crew group. The author focused on implementing Benders decomposition for effectively solving the model, where both the master program and the subproblem were solved using a branch-and-price strategy. In addition, *Pareto-optimal* or nondominated Benders cuts were generated using the technique of Magnanti and Wong (1981) to accelerate the solution process and to address degeneracy in the Benders subproblem. Papadakos (2009) tested his model on seven instances, all of which considered six aircraft fleets and a number of legs ranging from 214 to 705. Central processing unit (CPU) times ranging from 0.35 to 27.8 hours were reported for solving the proposed model to near-optimality.

There is also a rich body of literature addressing partial integrations of different suitable combinations of the airline scheduling stages. Notable in this respect, Clarke et al. (1996) incorporated considerations of maintenance opportunities and crew rests within the fleet assignment framework by introducing additional arcs that enforce certain timelines at specific stations. Barnhart et al. (1998a) proposed a model that incorporates fleet assignment and aircraft routing (FAAR) within a single framework by defining an *augmented flight string*, i.e., a sequence of connected flights that begin and end at maintenance stations with an additional minimum maintenance time appended to the end. Haouari, Aissaoui, and Mansour (2009); Mansour et al. (2010); and Haouari et al. (2011) also investigated combined aircraft fleeting and routing decisions along with consideration of *deadhead flights*, i.e., flights that serve to relocate aircraft and crews. Integration of FAAR is beneficial, especially for early-stage scheduling decisions where the complications of considering crew pairing decisions are temporarily put aside. Moreover, such a model can incorporate several useful features such as maintenance routing rules and consideration of optional flights. However, the resulting fleeting and routing decisions may be incompatible with crew pairing decisions that are made at a later stage.

Recognizing this perspective, researchers have also integrated aircraft routing with crew scheduling as the decisions from the former stage strongly impact the feasibility of the latter. Among related works, Cohn and Barnhart (2003) proposed an extended crew pairing model wherein the key aircraft routing decisions about short connections are delayed to the crew pairing stage. Cordeau et al. (2001); Mercier, Cordeau, and Soumis (2005); Mercier and Soumis (2007); and Mercier (2008) investigated different models and solution schemes for the integrated aircraft routing and crew pairing model. Integrating aircraft routing and crew pairing is valuable for making operationally optimal decisions relatively closer to the time of departure (usually 30–60 days prior). However, by itself, this does not have the flexibility of refleeting when encountering infeasible routing and pairing situations.

All of these integrated models are typically formulated using node-arc or set partitioning representations based on time-space and flight connection networks. Note that the size of such integrated models grows rapidly with the number of aircraft and flight legs, making it difficult to directly solve even moderately sized problems. Hence, these models are often implemented using sophisticated, specialized algorithms based on column generation and Benders decomposition techniques.

3. Model Formulation

Before presenting our integrated model, we introduce the following notation based on a *flight connection network* involving a set of daily flights:

Notation: Let L denote the set of daily flight legs under consideration. For each flight leg $j \in L$, we define the following notation (with all time durations expressed in minutes):

$DT_j \in [0, 1{,}440]$: departure time of flight leg j.
$AT_j \in [0, 1{,}440]$: arrival time of flight leg j.
DS_j: departure station of flight leg j.
AS_j: arrival station of flight leg j.
t_j: flying time of flight leg j (we assume that $t_j \le 1{,}440$ for practical reasons).
na: number of *wrap-around flights* in the schedule (i.e., flights that depart on a given day, cross the end of the time horizon, and arrive the next day).
$L_w \equiv \{j \in L$: j is a wrap-around flight$\}$. Note that $na \equiv |L_w|$ by definition.

Let K denote the set of all aircraft types. For each aircraft type, we define:

τ^k: turn-time for aircraft of type k, $\forall k \in K$.
NA^k: number of available aircraft of type k, $\forall k \in K$.
$t^k_{\max}$: maximum flying time between two consecutive maintenance checks for aircraft of type k (note that $t_j \le t^k_{\max}$, $\forall j \in L$, $k \in K$).
$to^k_{\max}$: maximum number of takeoffs between two consecutive maintenance checks for aircraft of type k ($to^k_{\max} \ge 1$, $\forall k \in K$).
$d^k_{\max}$: maximum number of days between two consecutive maintenance checks for aircraft of type k ($d^k_{\max} \ge 1$, $\forall k \in K$).
M^k: time duration of a maintenance check for aircraft of type k, $\forall k \in K$ ($\tau^k < M^k < 1{,}440$).
S^k: set of maintenance stations for aircraft of type k, $\forall k \in K$ ($|S^k| \ge 1$).

For a given flight schedule, we define an associated digraph $G = (V, A)$ wherein each node $j \in V$ represents a flight leg. Furthermore, for each aircraft of type $k \in K$, we define a corresponding arc set A^k, where $A \equiv \bigcup_{k\in K} A^k$, and where each arc $a \in A^k$ represents a feasible connection, i.e., an arc $a \in A^k$ if and only if an aircraft of type k can consecutively serve the flights pertaining to the from-node and the to-node of this arc, respectively denoted as a^- and a^+. Also, for notational convenience, we denote the set of arcs that are incident to, and outgoing from, node $j \in V$ by δ^-_j and δ^+_j, respectively. More specifically, following Haouari, Shao, and Sherali (2013), the set of arcs A^k, for each $k \in K$, is given by the union of six arc subsets A^k_1, A^k_2, A^k_3, A^k_4, A^k_5, and A^k_6 that are defined as follows:

- An arc $a \in A^k_1$ if and only if a maintenance check could be planned between the arrival of flight leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$; both flight legs must be served consecutively on the same day. Hence, $(j, l) \in A^k_1 \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \in S^k$; and (iii) $AT_j + M^k \le DT_l$.
- An arc $a \in A^k_2$ if and only if a maintenance check could be planned between the arrival of flight leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$, and the same aircraft must serve flight leg l the day after serving flight leg j, even if $AT_j + \tau^k \le DT_l$. Hence, $(j, l) \in A^k_2 \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \in S^k$; and (iii) $DT_l < AT_j + M^k \le DT_l + 1{,}440$.
- An arc $a \in A^k_3$ if and only if a maintenance check could not be planned between the arrival of flight leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$, and both flight legs must be served consecutively on the same day. Hence, $(j, l) \in A^k_3 \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \notin S^k$ or $DT_l < AT_j + M^k$; and (iii) $AT_j + \tau^k \le DT_l$.
- An arc $a \in A^k_4$ if and only if a maintenance check could not be planned between the arrival of flight leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$, and the same aircraft serves flight leg l the day after serving flight leg j. Hence, $(j, l) \in A^k_4 \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \notin S^k$ or $AT_j + M^k > DT_l + 1{,}440$; and (iii) $DT_l < AT_j + \tau^k \le DT_l + 1{,}440$.
- An arc $a \in A^k_5$ if and only if a maintenance check could be planned between the arrival of leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$, and the same aircraft must serve flight leg l two days after serving flight leg j. This type of arc represents the situation in which the aircraft does not have enough time to undergo maintenance after serving flight j and then to serve flight l the next day. Hence, $(j, l) \in A^k_5 \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \in S^k$; and (iii) $DT_l + 1{,}440 < AT_j + M^k \le DT_l + 2{,}880$.

- An arc $a \in A_6^k$ if and only if a maintenance check could not be planned between the arrival of flight leg $a^- \equiv j$ and the departure of flight leg $a^+ \equiv l$, and the same aircraft must serve flight leg l two days after serving flight leg j. Hence, $(j, l) \in A_6^k \Leftrightarrow$ (i) $AS_j \equiv DS_l$; (ii) $AS_j \notin S^k$; and (iii) $DT_l + 1{,}440 < AT_j + \tau^k \le DT_l + 2{,}880$.

Furthermore, to incorporate itinerary-based flight demands, we define the following notation:

Π: set of all itineraries.
$\Pi_j \subseteq \Pi$: the subset of itineraries that include flight j, $\forall j \in L$.
H: set of all fare classes.
γ_{kh}: capacity of aircraft of type $k \in K$ to accommodate passengers for fare class $h \in H$.
μ_{ph}: mean demand for fare class $h \in H$ within itinerary $p \in \Pi$.
f_{ph}: estimated revenue for fare class $h \in H$ within itinerary $p \in \Pi$.
r_a: estimated through-value for connection $a \in A$.

In addition, we adopt the same fleet assignment cost definition used in Mansour et al. (2010), which incorporates the fixed operating charges as well as the opportunity cost due to *spilled passengers*. Accordingly, we denote the cost associated with assigning an aircraft of type k to flight leg j as c_{jk}, which can be expressed as

$$c_{jk} = \bar{c}_{jk} + \sum_{h\in H} o_{jh}\left(\sum_{p\in\Pi_j} \mu_{ph} - \gamma_{kh}\right)^+, \tag{1}$$

where $\bar{c}_{jk}$ denotes the fixed cost of assigning fleet type k to flight j, and o_{jh} represents the opportunity cost per *spilled passenger* on flight j incurred where the expected demand for fare class h exceeds the capacity of the assigned aircraft, and where $(\cdot)^+ \equiv \max\{0, \cdot\}$.

Decision Variables:

x_a: binary variable that equals 1 if arc $a \in A$ is selected, and 0 otherwise.
y_{jk}: binary variable that equals 1 if flight leg $j \in L$ is assigned to an aircraft of type k, and 0 otherwise.
u_j^k: total accumulated flying hours for aircraft of type $k \in K$ since its last maintenance check after serving flight leg $j \in L$.
v_j^k: total number of takeoffs for aircraft of type $k \in K$ since its last maintenance check after serving flight leg $j \in L$.
d_j^k: total number of elapsed days for aircraft of type $k \in K$ since the last maintenance check after serving flight leg $j \in L$.
π_{ph}: number of passengers accepted for itinerary $p \in \Pi$ within fare class $h \in H$.

Based on the above notation, we first formulate an integrated *fleet assignment and aircraft routing* (**FAAR**) problem as follows, where for convenience, we denote the set of maintenance-permitting arcs as $A_M^k \equiv A_1^k \cup A_2^k \cup A_5^k$, and the set of nonmaintenance arcs as $A_{NM}^k \equiv A_3^k \cup A_4^k \cup A_6^k$, for each $k \in K$:

FAAR:

$$\text{maximize} \left\{\sum_{p\in\Pi}\sum_{h\in H} f_{ph}\pi_{ph} + \sum_{a\in A} r_a x_a - \sum_{j\in L}\sum_{k\in K} c_{jk}y_{jk}\right\} \tag{2}$$

subject to

$$\sum_{k\in K} y_{jk} = 1, \quad \forall j \in L, \tag{3}$$

$$\sum_{a\in\delta_j^-\cap A^k} x_a = y_{jk}, \quad \forall j \in L, k \in K, \tag{4}$$

$$\sum_{a\in\delta_j^+\cap A^k} x_a = y_{jk}, \quad \forall j \in L, k \in K, \tag{5}$$

$$u_j^k x_a = t_j x_a, \quad \forall a \in \delta_j^- \cap A_M^k, j \in L, k \in K, \tag{6}$$

$$u_j^k x_a = (u_{a^-}^k + t_j)x_a, \quad \forall a \in \delta_j^- \cap A_{NM}^k, j \in L, k \in K, \tag{7}$$

$$v_j^k x_a = x_a, \quad \forall a \in \delta_j^- \cap A_M^k, j \in L, k \in K, \tag{8}$$

$$v_j^k x_a = (v_{a^-}^k + 1)x_a, \quad \forall a \in \delta_j^- \cap A_{NM}^k, j \in L, k \in K, \tag{9}$$

$$d_j^k x_a = x_a, \quad \forall a \in \delta_j^- \cap A_M^k, j \in L, k \in K, \tag{10}$$

$$d_j^k x_a = d_{a^-}^k x_a, \quad \forall a \in \delta_j^- \cap A_3^k, j \in L, k \in K, \tag{11}$$

$$d_j^k x_a = (d_{a^-}^k + 1)x_a, \quad \forall a \in \delta_j^- \cap A_4^k, j \in L, k \in K, \tag{12}$$

$$d_j^k x_a = (d_{a^-}^k + 2)x_a, \quad \forall a \in \delta_j^- \cap A_6^k, j \in L, k \in K, \tag{13}$$

$$\sum_{a\in A_2^k\cup A_4^k} x_a + 2\sum_{a\in A_5^k\cup A_6^k} x_a + \sum_{j\in L_w} y_{jk} \le NA^k, \quad \forall k \in K, \tag{14}$$

$$t_j \le u_j^k \le t_{\max}^k, \quad \forall j \in L, k \in K, \tag{15}$$

$$1 \le v_j^k \le to_{\max}^k, \quad \forall j \in L, k \in K, \tag{16}$$

$$1 \le d_j^k \le d_{\max}^k, \quad \forall j \in L, k \in K, \tag{17}$$

$$\sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K} \gamma_{kh} y_{jk}, \quad \forall j \in L, h \in H, \tag{18}$$

$$0 \le \pi_{ph} \le \mu_{ph}, \quad \forall p \in \Pi, h \in H, \tag{19}$$

$$(x, y) \text{ binary.} \tag{20}$$

The objective function (2) maximizes the overall profit given by the total revenue, including through-values, minus the fleet assignment costs. Constraint (3) requires each flight leg to be covered by exactly one fleet type. Constraints (4) and (5) ensure that each flight has exactly one predecessor and one successor, respectively, both of which are assigned to the same aircraft type. Hence, together with x being binary-valued, these restrictions induce the solution to be composed of cycles or cyclic rotations, where each rotation covers a set of flights using a particular type of aircraft. Note that since each rotation describes a cyclic use of a particular aircraft of some type, the traditional conservation of flow constraints at stations written for typical fleet assignment models using a time-space network representation are automatically satisfied (see, for example, Sherali, Bish, and Zhu 2006). For each aircraft type, the nonlinear constraints (6)–(7) together with (15) enforce the total flying time restrictions. Note that the nature of these cyclic constraints precludes a rotation having no maintenance visit. Similarly, Constraints (8)–(9) and (16) ensure the restrictions on the maximal number of takeoffs, and Constraints (10)–(13) and (17) likewise guarantee restrictions on the maximum number of days between maintenance checks. Observe that, given binary values of (x, y) feasible to Constraints (3)–(5), Constraints (8)–(13) automatically induce the v- and d-variables to be integer-valued; hence, these variables are logically declared in Constraints (16) and (17) to be simply continuous-valued. (In this same vein, if t_j are integer-valued, then so are the u-variable values in any feasible solution.) Constraint (14) requires that the total number of aircraft in service not exceed the available size of the fleet (where $\sum_{k\in K}\sum_{j\in L_w} y_{jk} = |L_w| = na$ by (3)). Note that this is similar to an aircraft count constraint used in fleet assignment models (e.g., see Sherali, Bish, and Zhu 2006), where the accounting in this case is done at the count timeline $t = 1{,}440$, and where the first two terms in the left-hand side (LHS) of (14) count the number of aircraft of type k on the ground, and where the third term equals the number of aircraft of type k in the air at that time. Constraint (18) guarantees that the total number of passengers traveling on a flight within a specific fare class is no more than the capacity available for that fare class on the assigned aircraft. Constraint (19) requires that the number of passengers accepted on any particular itinerary for each fare class not exceed the corresponding expected demand. Finally, Constraint (20) imposes logical restrictions on the binary decision variables.

For the sake of simplicity, we adopt a basic formulation of the passenger-mix via Constraints (18)–(19), wherein the possibility of recapturing spilled passengers among fare classes and between itineraries are not considered. This may underestimate the true passenger revenue, since in reality a proportion of spilled passengers can be re-accommodated by other means. To overcome this drawback, recapture-based constraints can be incorporated within the model as in Sherali, Bae, and Haouari (2013), for example, to redistribute a proportion of spilled passengers. Alternatively, a more sophisticated representation of the passenger-mix problem can be accommodated following Barnhart, Kniker, and Lohatepanont (2002). This would enhance model accuracy but at the expense of increasing its complexity. We relegate such considerations for future research.

Moreover, the daily demand considered serves as a rough estimate for day-to-day passenger flows, which may not accurately reflect actual daily traffic. Note also the complexity of predicting demand and correspondingly effecting pricing strategies. Additionally, note that when we examine the quality of a solution, we evaluate it using the best mix of passengers from the viewpoint of optimizing the overall objective function (i.e., we adopt a systems perspective). However, in reality, passengers make choices sequentially, and those making early reservations at lower prices may consume seating capacity in lieu of more profitable customers seeking reservations at a later stage. Such user preferences and associated revenue management issues are neglected in the model, and we recommend their consideration for future research. That being said, the passenger-mix model used here is designed to provide a simple yet reasonable approximation for solving early-stage scheduling problems when reservation details are yet to be ascertained. However, in the context of flight network optimization, the aforementioned drawbacks can be somewhat mitigated by intentionally planning for a lower load factor, thus reserving seats for close-in (and thus more valuable) bookings. Nevertheless, to comprehensively address the booking yields of an airline company, a *revenue management* system must be implemented; we leave this for future research.

Following Sherali, Bae, and Haouari (2010), we can replace (18) by the tighter inequality

$$\sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K} \tilde{\gamma}_{jkh} y_{jk}, \quad \forall\, j\in L,\ h\in H, \tag{21}$$

where $\tilde{\gamma}_{jkh} \equiv \min\{\gamma_{kh}, \sum_{p\in\Pi_j}\mu_{ph}\}$, $\forall\, j\in L,\ k\in K,\ h\in H$.

Moreover, we can replace (19) by

$$0 \le \pi_{ph} \le \tilde{\mu}_{ph} \equiv \min\{\mu_{ph}, \max_{k\in K}\gamma_{kh}\}, \quad \forall p\in\Pi, h\in H. \tag{22}$$

Remark 1. The y-variables can be eliminated from the formulation by substituting them in terms of the x-variables using Constraint (4) or (5). Practically, to induce sparsity in the resulting model, we can substitute y_{jk} with either of the expressions on the LHS of (4) and (5), whichever yields an overall sparser constraint (3), for each $j \in L$. Arbitrarily selecting (4) in this context, Constraints (3)–(5), (14), and (21) can be replaced by the following:

$$\sum_{a\in\delta_j^-} x_a = 1, \quad \forall\, j \in L, \tag{23}$$

$$\sum_{a\in\delta_j^-\cap A^k} x_a = \sum_{a\in\delta_j^+\cap A^k} x_a, \quad \forall\, j \in L,\ k \in K, \tag{24}$$

$$\sum_{a\in A_2^k\cup A_4^k} x_a + 2\sum_{a\in A_5^k\cup A_6^k} x_a + \sum_{j\in L_w}\sum_{a\in\delta_j^-\cap A^k} x_a \le NA^k, \quad \forall\, k \in K, \tag{25}$$

$$\sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K}\sum_{a\in\delta_j^-\cap A^k} \tilde{\gamma}_{jkh} x_a, \quad \forall\, j \in L,\ h \in H, \tag{26}$$

and the objective function can be rewritten as follows:

$$\text{maximize}\left\{\sum_{p\in\Pi}\sum_{h\in H} f_{ph}\pi_{ph} - \sum_{j\in L}\sum_{k\in K}\sum_{a\in\delta_j^-\cap A^k} (c_{jk} - r_a)x_a\right\}. \quad \square \tag{27}$$

3.1. Reformulation and Linearization

To enhance the solvability of the integrated model, we next propose to apply the special-structured Reformulation-Linearization Technique (RLT) of Sherali, Adams, and Driscoll (1998) to derive a tight, equivalent linear model representation of this problem. Toward this end, first consider Constraints (6) and (7). Following Haouari, Shao, and Sherali (2013), we linearize these constraints using the substitutions:

$$\omega_{jl}^k = u_j^k x_{jl}, \quad \forall\, (j,l) \in A^k,\ k \in K, \tag{28}$$

$$\rho_{jl}^k = u_l^k x_{jl}, \quad \forall\, (j,l) \in A^k,\ k \in K. \tag{29}$$

Accordingly, Constraints (6) and (7) become

$$\rho_{jl}^k = t_l x_{jl}, \quad \forall\, (j,l) \in A_M^k,\ k \in K, \tag{30}$$

$$\rho_{jl}^k = \omega_{jl}^k + t_l x_{jl}, \quad \forall\, (j,l) \in A_{NM}^k,\ k \in K. \tag{31}$$

To ensure the product relationships (28) and (29) within the RLT process, consider the following set SS_j^k, $\forall\, j \in L$, $k \in K$, implied by (20), (23), and (24):

$$SS_j^k = \left\{ x\colon \begin{array}{l} \displaystyle\sum_{a\in\delta_j^-\cap A^k} x_a = \sum_{a\in\delta_j^+\cap A^k} x_a \\ \displaystyle\sum_{a\in\delta_j^-\cap A^k} x_a \le 1 \\ x_a \ge 0, \quad \forall\, a \in (\delta_j^- \cup \delta_j^+) \cap A^k \end{array} \right\}. \tag{32}$$

Since SS_j^k implies $\{0 \le x_a \le 1,\ \forall\, a \in (\delta_j^- \cup \delta_j^+) \cap A^k\}$, $\forall\, j \in L$, $k \in K$, it suffices to take interproducts of the constraints defining SS_j^k with (15) for each $j \in L, k \in K$, to enforce (28) and (29) while deriving a tight linear programming representation (see Sherali, Adams, and Driscoll 1998). Hence, for each $j \in L, k \in K$, multiplying Constraint (15) by x_{lj}, $\forall\, (l,j) \in A^k$, and by x_{jl}, $\forall\, (j,l) \in A^k$, we respectively obtain the following inequalities on linearization (where we have interchanged the indices l and j for the first case):

$$t_l x_{jl} \le \rho_{jl}^k \le t_{\max}^k x_{jl}, \quad \forall\, (j,l) \in A^k,\ k \in K, \tag{33}$$

$$t_j x_{jl} \le \omega_{jl}^k \le t_{\max}^k x_{jl}, \quad \forall\, (j,l) \in A^k,\ k \in K. \tag{34}$$

Next, multiplying the first constraint of (32) in SS_j^k by simply u_j^k (noting that this is an equality restriction), we derive the following constraint on linearization using (28) and (29):

$$\sum_{l:\,(l,j)\in A^k} \rho_{lj}^k = \sum_{l:\,(j,l)\in A^k} \omega_{jl}^k, \quad \forall\, j \in L,\ k \in K. \tag{35}$$

Finally, multiplying the second constraint of (32) in SS_j^k with the two bound-factors in (15) lifts the latter (noting (33)) to the following restrictions on linearization using (28) and (29):

$$t_j + \sum_{l:\,(l,j)\in A^k} (\rho_{lj}^k - t_j x_{lj}) \le u_j^k \le t_{\max}^k - \sum_{l:\,(l,j)\in A^k} (t_{\max}^k x_{lj} - \rho_{lj}^k), \quad \forall\, j \in L,\ k \in K. \tag{36}$$

Thus, by Sherali, Adams, and Driscoll (1998), we can derive an equivalent lifted model by replacing (6), (7), and (15) in the problem with (30)–(36). However, observe that the variable u_j^k appears only in (36) within the resulting model. So this constraint can be eliminated and the u-variables can be determined posteriori using (36) so long as

$$t_j + \sum_{l:\,(l,j)\in A^k} (\rho_{lj}^k - t_j x_{lj}) \le t_{\max}^k - \sum_{l:\,(l,j)\in A^k} (t_{\max}^k x_{lj} - \rho_{lj}^k), \quad \forall\, j \in L,\ k \in K, \quad \text{i.e.,}$$

$$t_j\left(1 - \sum_{l:\,(l,j)\in A^k} x_{lj}\right) \le t_{\max}^k\left(1 - \sum_{l:\,(l,j)\in A^k} x_{lj}\right), \quad \forall\, j \in L, k \in K,$$

which automatically holds because of $t_j \le t_{\max}^k$ and $\sum_{l:\,(l,j)\in A^k} x_{lj} \le 1$ by (23). Thus, we can drop (36) and replace (6), (7), and (15) with (30)–(35).

By using (30) and (31) to further eliminate the ρ-variables from (30)–(35), we readily establish the following proposition (which is similar to Proposition 2 in Haouari, Shao, and Sherali 2013).

Proposition 1. *Constraints* (30)–(35) *can be replaced by the restrictions* (37)–(38), *which, together with* (23) *and* (24), *yield an equivalent set of constraints, even in the sense of the continuous linear program* (LP) *relaxation*

$$\sum_{l:\,(j,l)\in A^k} \omega_{jl}^k = t_j \sum_{l:\,(l,j)\in A^k} x_{lj} + \sum_{l:\,(l,j)\in A_{NM}^k} \omega_{lj}^k, \quad \forall\, j \in L, k \in K, \tag{37}$$

$$t_j x_{jl} \le \omega_{jl}^k \le b_{jl}^{tk} x_{jl}, \quad \forall\, (j,l) \in A^k, k \in K, \tag{38}$$

where $b_{jl}^{tk} = \begin{cases} t_{\max}^k - t_l, & (j,l) \in A_{NM}^k, \\ t_{\max}^k, & (j,l) \in A_M^k. \end{cases}$ $\square$

The nonlinear constraints (8)–(13) can be lifted and linearized following an identical derivation by defining the new set of RLT variables

$$\eta_{jl}^k = v_j^k x_{jl}, \quad \forall (j,l) \in A^k,\ k \in K, \tag{39}$$

$$\lambda_{jl}^k = d_j^k x_{jl}, \quad \forall (j,l) \in A^k,\ k \in K, \tag{40}$$

and by introducing the following parameters for each $k \in K$, similar to the parameters $b_a^{t^k}$, $a \in A$, defined in Proposition 1 (written using the notation $a \in A^k$ in lieu of $(j,l) \in A^k$):

$$b_a^{to^k} = \begin{cases} to_{\max}^k - 1, & a \in A_{NM}^k, \\ to_{\max}^k, & a \in A_M^k, \end{cases}$$

$$b_a^{d^k} = \begin{cases} d_{\max}^k - 1, & a \in A_4^k, \\ d_{\max}^k - 2, & a \in A_6^k, \\ d_{\max}^k, & a \in A_M^k \cup A_3^k. \end{cases}$$

This yields the following equivalent reformulation of Problem FAAR:

FAAR-RLT:

$$\text{maximize} \left\{ \sum_{p\in\Pi}\sum_{h\in H} f_{ph}\pi_{ph} - \sum_{j\in L}\sum_{k\in K}\sum_{a\in\delta_j^-\cap A^k}(c_{jk}-r_a)x_a \right\} \tag{41}$$

$$\text{subject to} \quad \sum_{a\in\delta_j^-} x_a = 1, \quad \forall j \in L, \tag{42}$$

$$\sum_{a\in\delta_j^-\cap A^k} x_a = \sum_{a\in\delta_j^+\cap A^k} x_a, \quad \forall j\in L,\ k\in K, \tag{43}$$

$$\sum_{a\in\delta_j^+\cap A^k} \omega_a^k = t_j \sum_{a\in\delta_j^-\cap A^k} x_a + \sum_{a\in\delta_j^-\cap A_{NM}^k} \omega_a^k, \quad \forall j\in L,\ k\in K, \tag{44}$$

$$t_{a^-} x_a \le \omega_a^k \le b_a^{t^k} x_a, \quad \forall a\in A^k,\ k\in K, \tag{45}$$

$$\sum_{a\in\delta_j^+\cap A^k} \eta_a^k = \sum_{a\in\delta_j^-\cap A^k} x_a + \sum_{a\in\delta_j^-\cap A_{NM}^k} \eta_a^k, \quad \forall j\in L,\ k\in K, \tag{46}$$

$$x_a \le \eta_a^k \le b_a^{to^k} x_a, \quad \forall a\in A^k,\ k\in K, \tag{47}$$

$$\sum_{a\in\delta_j^+\cap A^k} \lambda_a^k = \sum_{a\in\delta_j^-\cap A^k} x_a + \sum_{a\in\delta_j^-\cap A_6^k} x_a - \sum_{a\in\delta_j^-\cap A_3^k} x_a + \sum_{a\in\delta_j^-\cap A_{NM}^k} \lambda_a^k, \quad \forall j\in L, k\in K, \tag{48}$$

$$x_a \le \lambda_a^k \le b_a^{d^k} x_a, \quad \forall a\in A^k,\ k\in K, \tag{49}$$

$$\sum_{a\in A_2^k\cup A_4^k} x_a + 2\sum_{a\in A_5^k\cup A_6^k} x_a + \sum_{j\in L_w}\sum_{a\in\delta_j^-\cap A^k} x_a \le NA^k, \quad \forall k\in K, \tag{50}$$

$$\sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K}\sum_{a\in\delta_j^-\cap A^k} \tilde{\gamma}_{jkh} x_a, \quad \forall j\in L,\ h\in H, \tag{51}$$

$$0 \le \pi_{ph} \le \tilde{\mu}_{ph}, \quad \forall p\in\Pi,\ h\in H, \tag{52}$$

$$x \text{ binary}, \tag{53}$$

where we have used (23)–(27) to rewrite the objective function and Constraints (3)–(5), (14), and (21), and where we have rewritten (37) and (38) of Proposition 1 in more compact form as (44) and (45), respectively, with (46)–(49) being a similar representation derived for (8)–(13) and (16)–(17).

3.1.1. Lower Bound on the Number of Required Aircraft. In this section, we briefly introduce a valid inequality on the total number of aircraft. Given the flight schedule, suppose that we extend the duration of all flights by the minimum turn-time among all fleet types (denoted by $\tau_{\min}$), and lay out these resulting flight durations on a daily timeline. Then the number of aircraft required must be equal to or exceed the maximum number of overlapping flight intervals at any point in time. Denoting this value by Γ, it follows that

$$\sum_{k\in K}\left(\sum_{a\in A_2^k\cup A_4^k} x_a + 2\sum_{a\in A_5^k\cup A_6^k} x_a + \sum_{j\in L_w}\sum_{a\in\delta_j^-\cap A^k} x_a \right) \ge \Gamma. \tag{54}$$

In addition, similar to Haouari, Shao, and Sherali (2013), another lower bound can be obtained by minimizing the LHS of Constraint (54) with regard to the LP relaxation of Constraints (42)–(53). Denoting the optimal objective value of this program by Γ^*, the right-hand-side (RHS) of (54) can therefore be replaced with $\max\{\Gamma, \Gamma^*\}$.

3.2. Further Integration with Crew Pairing

We next propose a further integrated airline operational model by extending Model FAAR-RLT to include crew pairing decisions. Crew pairings account for a significant proportion of airline operational costs, and are intimately intertwined with FAAR decisions. Because it is generally difficult to handle these decisions directly by using path-based restrictions along with the accompanying nonlinear crew cost components in a polynomially-sized, but large-scale, modeling framework, we implicitly generate sets of feasible crew pairings and adopt the traditional set partitioning formulation to incorporate crew pairing aspects in Model FAAR-RLT. Toward this end, consider the following additional notation:

- F: set of aircraft families, indexed by f.
- K^f: set of types of aircraft that belong to family f. (Note that $K = \bigcup_{f\in F} K^f$.)
- P^f: set of admissible crew pairings that can serve flights covered by aircraft of family f.
- S^f: set of short connections, indexed by σ, which can be served by aircraft of family f.

$\mathcal{A}^f_\sigma$: set of arcs $a \in \bigcup_{k\in K^f} A^k$ between the associated pair of flights that correspond to the short connection $\sigma \in S^f$.

e^f_{jp}: binary indicator that equals 1 if flight leg j is covered by pairing $p \in P^f$ of aircraft family f, and 0 otherwise.

$s^f_{\sigma p}$: binary indicator that equals 1 if short connection $\sigma \in S^f$ is covered by pairing $p \in P^f$ of aircraft family f, and 0 otherwise.

w^f_p: cost associated with pairing $p \in P^f$ of aircraft family f.

Furthermore, we introduce the following additional binary decision variable pertaining to pairings:

z^f_p: binary variable that equals 1 if pairing $p \in P^f$ of aircraft family f is selected, and 0 otherwise.

The resulting integrated fleet assignment, aircraft routing, and crew pairing model (**FRC**) can then be formulated as follows:

FRC:

$$\text{maximize } \left\{ \sum_{p\in\Pi}\sum_{h\in H} f_{ph}\pi_{ph} - \sum_{j\in L}\sum_{k\in K}\sum_{a\in\delta^-_j\cap A^k} (c_{jk}-r_a)x_a - \sum_{f\in F}\sum_{p\in P^f} w^f_p z^f_p \right\} \quad (55)$$

subject to (42)–(53) along with

$$\sum_{p\in P^f} e^f_{jp} z^f_p = \sum_{k\in K^f}\sum_{a\in\delta^-_j\cap A^k} x_a, \quad \forall j \in L, f \in F, \quad (56)$$

$$\sum_{p\in P^f} s^f_{\sigma p} z^f_p \le \sum_{a\in\mathcal{A}^f_\sigma} x_a, \quad \forall \sigma \in S^f, f \in F, \quad (57)$$

$$z \text{ binary.} \quad (58)$$

The objective function (55) incorporates an additional crew pairing cost term in (41). Constraint (56) requires that each flight be served by a crew that is eligible for the specific aircraft family assigned to this flight, where the RHS of (56) equals $\sum_{k\in K^f} y_{jk}$. Constraint (57) imposes restrictions on permissible short connections, i.e., a crew pairing pertaining to aircraft family $f \in F$ can cover a short connection $\sigma \in S^f$ only if the two consecutive flights associated with the short connection σ are served by an aircraft from the same family f. Finally, Constraint (58) represents the logical binary restrictions on the crew pairing decision variables.

4. Solution Methodology

U.S. airline companies usually operate a daily schedule of hundreds of domestic flights with a variety of large-sized fleets. Moreover, a typical airline that uses a hub-and-spoke flight network can hold thousands of itineraries in its profile, and millions of feasible aircraft rotations as well as crew pairings. For such large-sized airlines, our modeling framework can focus on a relatively smaller fleet of wide-body aircraft, along with major associated airports and a related manageable set of principal itineraries to ensure tractability. On the other hand, there are also several small airlines that maintain a moderate fleet of aircraft and operate only a few flights in their network, for which a full-scale model is practically implementable. Even so, the integrated model inherits a huge number of variables and constraints that preclude a direct solution using off-the-shelf software. It is therefore imperative to design a specialized decomposition-based solution methodology. This motivates us to propose a sequential-fixing approach that uses Benders decomposition for solving the integrated Model FRC. In particular, the constraints (51)–(52) pertaining to itinerary-based demands, and the crew pairing restrictions (56)–(58), are handled in separate subproblems in the Benders decomposition framework, where the generated cuts therefrom are appended to the aircraft fleeting and routing master program that is composed of Constraints (42)–(50) and (53)–(54).

We first solve Model FAAR-RLT in isolation and, subsequently, solve the CP with the fixed fleeting and aircraft routing decisions. The net solution obtained from these two sequential stages serves as a benchmark for the upcoming integrated model solution, where the optimal crew pairings at hand are also used to initialize the column generation procedure in the corresponding crew pairing integrated model subproblem. In the next phase, we turn to Model FRC and break it down using a Benders partitioning approach. This leads to an itinerary-based passenger-mix subproblem and a crew pairing subproblem, where the Benders relaxed master program (RMP) addresses aircraft fleeting and routing decisions. In this process, we generate maximal nondominated Benders cuts as proposed by Sherali and Lunday (2013), and we adopt a set of initial cuts for accelerating the solution procedure as recommended by McDaniel and Devine (1977). For the crew pairing subproblem, we use a perturbed Lagrangian dual strategy along with a deflected subgradient optimization technique as proposed by Subramanian and Sherali (2008) to effectively generate useful columns that avoid stalling of the underlying column generation procedure. We embed this in a branch-and-price heuristic (see Barnhart et al. 1998b and Mercier, Cordeau, and Soumis 2005) while adopting the *branch-on-the-follow-on* strategy (Ryan and Foster 1981) for selecting branching variables. The detailed steps of this overall procedure are described next.

4.1. Itinerary-Based Passenger-Mix Subproblem

For each fare class $h \in H$, the model yields the following itinerary-based passenger-mix subproblem (IPM_h):

IPM_h:

$$\text{maximize} \quad \sum_{p\in\Pi} f_{ph}\pi_{ph} \tag{59}$$

$$\text{subject to} \quad \sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K} \sum_{a\in\delta_j^-\cap A^k} \tilde{\gamma}_{jkh}\bar{x}_a, \quad \forall\, j \in L, \tag{60}$$

$$0 \le \pi_{ph} \le \tilde{\mu}_{ph}, \quad \forall\, p \in \Pi, \tag{61}$$

where $\bar{x}$ is the solution inherited from the Benders RMP.

For each $h \in H$, let χ^1_{jh}, $j \in L$, and χ^2_{ph}, $p \in \Pi$, denote the dual variables associated with (60) and the upper-bounding constraints in (61), respectively. Let X_h denote an appropriate restricted subset of extreme points of the dual feasible region corresponding to Problem IPM_h. Because Problem IPM_h has complete recourse, i.e., it is feasible and bounded for any given (nonnegative) solution $\bar{x}$, the optimal value function ϕ_h for Problem IPM_h can therefore be characterized by the following set of Benders optimality cuts:

$$\phi_h \le \sum_{j\in L}\sum_{k\in K}\sum_{a\in\delta_j^-\cap A^k} \chi^1_{jh}\tilde{\gamma}_{jkh}x_a + \sum_{p\in\Pi}\chi^2_{ph}\tilde{\mu}_{ph}, \quad \forall\,(\chi^1_{jh}, \chi^2_{ph}) \in X_h,\; h \in H. \tag{62}$$

4.2. Crew Pairing Subproblem

The crew pairing subproblem, which is solved separately for each aircraft family $f \in F$, seeks to generate sequences of flights for crews to serve while satisfying a series of FAA-mandated work-rule restrictions. The objective is to find a set of feasible pairings that partitions the given aircraft family's flight network at a minimum cost. Specifically, for each $f \in F$, we examine the following continuous relaxation of the CP:

CP^f:

$$\text{minimize} \quad \left\{\sum_{p\in P^f} w^f_p z^f_p + \beta\sum_{j\in L} q^f_j\right\} \tag{63}$$

$$\text{subject to} \quad \sum_{p\in P^f} e^f_{jp} z^f_p + q^f_j = \sum_{k\in K^f}\sum_{a\in\delta_j^-\cap A^k} \bar{x}_a, \quad \forall\, j\in L, \tag{64}$$

$$\sum_{p\in P^f} s^f_{\sigma p} z^f_p + q^f_\sigma = \sum_{a\in\mathcal{A}^f_\sigma} \bar{x}_a, \quad \forall\,\sigma\in S^f, \tag{65}$$

$$(z, q) \ge 0, \tag{66}$$

where q^f_j, $\forall\, j \in L$, are artificial variables appended to Constraint (56) along with a penalty term in the objective function using a sufficiently large penalty parameter β that is taken to be an upper bound on possible crew pairing costs, and where q^f_σ, $\forall\,\sigma \in S^f$ are slack variables for Constraint (57). These artificial variables ensure that Problem CP^f has an optimal solution for any fixed $\bar{x}$ obtained from the master program, and thus enables us to characterize the corresponding optimal value function in the master program via only Benders optimality cuts. Note that this strategy of using artificial variables is motivated by the convenience of using a standard off-the-shelf software while ensuring complete recourse and thus avoiding infeasible subproblems. Nonetheless, such derived cuts can be translated into feasibility cuts if desired (Sherali and Lunday 2013).

Furthermore, note that the RHS values in (64)–(65) are 0 or 1 whenever $\bar{x}$ is binary-valued, which accordingly determine the set of flights that need to be covered by the crews trained to serve on aircraft of family f, including permissible short connections between designated consecutive pairs of flights. Note also that, in the sequel when we consider the continuous relaxation of the Benders master program, we might inherit fractional RHS values in Constraints (64)–(65). In this case, we consider all of the flights corresponding to positive RHS values in (64)–(65) within the network for solving the crew pairing subproblem. Observe that this would still yield valid Benders cuts.

Because of the nonlinear cost structure as well as the complexity of the pairing restrictions, Problem CP^f is usually solved using column generation, wherein the pricing subproblem is evaluated using a *constrained shortest path* (or *multilabel shortest path*) algorithm as first proposed by Desrochers and Soumis (1989). Theoretically, the constrained shortest path problem is NP-hard, and could enumerate all pairings for the network before termination; however, if at any node, the label values updated along one potential pairing are all better (smaller) than those for another pairing, then we say that the former pairing *dominates* the latter, where the dominated pairing can be eliminated from the solution pool. This feature greatly helps to reduce the solution effort. This solution process is terminated when either the optimality gap is reduced to within a specified threshold or a predefined number of iterations has been performed. See Shao (2013) for a detailed description of this procedure.

In addition, we note a recent attempt by Ahmad-Beygi, Cohn, and Weir (2009) to model the CP using a polynomially-sized (compact) integer programming formulation that can be directly solved using off-the-shelf software. A similar formulation styled in the fashion of the aircraft routing model proposed by Haouari, Shao, and Sherali (2013) is possible for this problem, which could be beneficially lifted using an RLT-based partial convexification process (see Sherali, Adams, and Driscoll 1998). Such compact formulations could instead be used to solve the subproblem for generating crew pairings. We leave the investigation of this strategy to future research.

The solution to the LP relaxation of the CP for each $f \in F$ provides a Benders optimality cut for the master program. In particular, for each $f \in F$, let ψ^1_{jf}, $\forall j \in L$, and $\psi^2_{\sigma f}$, $\forall \sigma \in S^f$, denote the dual variables associated with Constraints (64) and (65), respectively, and let Ψ^f denote an appropriate restricted subset of extreme points of the dual feasible region corresponding to Problem CP^f. Again, because Problem CP^f has complete recourse, the optimal value function ξ_f for Problem CP^f can be characterized within the master program via the following set of Benders optimality cuts:

$$\xi_f \geq \sum_{j \in L} \sum_{k \in K^f} \sum_{a \in \delta_j^- \cap A^k} \psi^1_{jf} x_a + \sum_{\sigma \in S^f} \sum_{a \in \mathcal{A}^f_\sigma} \psi^2_{\sigma f} x_a, \quad \forall (\psi^1_{jf}, \psi^2_{\sigma f}) \in \Psi_f,\ f \in F. \tag{67}$$

The column generation approach used for the CP often stalls when still remote from optimality, resulting in dual variable values that help little in generating improving pairings via the subproblem. To avoid such a phenomenon, which is identified as *dual noise* by Subramanian and Sherali (2008), we follow the authors' approach by solving a perturbed Lagrangian dual of the restricted crew pairing master program using a deflected subgradient optimization technique. This process has been shown to derive judicious low-norm dual solutions that generate beneficial columns.

Finally, to eventually obtain an integral solution to the crew pairing subproblem, we adopt a branch-and-price heuristic, wherein a *depth-first search* (DFS) tree is recursively used. At each node of the DFS tree, the associated LP relaxation is solved using the constrained shortest path algorithm. When a nonintegral solution yields an objective value that is smaller than the current upper bound, we apply the *branching-on-the-follow-on* rule (Ryan and Foster 1981) to create two subnodes in the DFS tree for further exploration.

4.3. Accelerating the Benders Decomposition Procedure

The Benders RMP is composed of the aircraft fleeting and routing decisions and constraints, along with Benders cuts obtained from the passenger-mix and the crew pairing subproblems. It is given as follows:

RMP:

$$\text{maximize} \quad \left\{ \sum_{h \in H} \phi_h - \sum_{j \in L} \sum_{k \in K} \sum_{a \in \delta_j^- \cap A^k} (c_{jk} - r_a) x_a - \sum_{f \in F} \xi_f \right\} \tag{68}$$

subject to (42)–(50), (53), (62), and (67).

We use the implementation scheme advocated by McDaniel and Devine (1977) to accelerate the solution procedure by initially relaxing the integrality restrictions in the master program and thus solving the continuous relaxation of Problem FRC via the Benders decomposition method. If the lower and upper bounds obtained respectively from the RMP and the Benders subproblem are sufficiently close, or some specified limit of iterations is reached, then we re-enforce the integrality restrictions on the x-variables as noted below, and continue the solution of the resulting mixed-integer program (MIP) using the implementation of the Benders procedure as recommended by Geoffrion and Graves (1974). In this process, rather than solve each current RMP to optimality as an MIP, we essentially solve just the full master program itself (denoted by MP) as an MIP, which is equivalent to solving the original problem FRC. This is accomplished by solving Problem MP using a branch-and-cut (B&C) approach, where Benders cuts are progressively generated as and when needed. Specifically, in this B&C algorithm, upper bounds are computed by solving the LP relaxation of the current RMP. Now, suppose that an integer-feasible solution is detected to RMP whose objective value is greater than the current lower bound (incumbent) value plus a prescribed threshold $\varepsilon_{RMP} \geq 0$. In this case, we pause and solve the Benders passenger-mix and crew pairing subproblems to evaluate the true objective function value of this solution in Problem MP (and therefore in Problem FRC) to update the incumbent solution to MP and generate new Benders cuts as necessary. We then continue the B&C algorithm. When the B&C algorithm discovers that no such integer-feasible solution exists, we terminate the overall algorithm with the indication that the incumbent solution is ε_{RMP}-optimal to the original problem.

To accelerate the convergence of the Benders decomposition method, Magnanti and Wong (1981) presented a seminal work for generating *nondominated* or *Pareto-optimal* Benders cuts using a core point selected in the relative interior of the convex hull of the primal feasible region defining the master program. Such nondominated cuts are particularly useful when the primal subproblem is highly degenerate, which induces a variety of possible Benders cuts at any iteration based on alternative optimal dual solutions. However, as demonstrated by Mercier and Soumis (2007) and Sherali and Lunday (2013), although the Magnanti-Wong method reduces the total number of generated cuts, it may not lead to a net advantage in computational effort because it requires solving an additional LP at each iteration. In addition, Rei et al. (2009) proposed an approach for generating multiple Benders and/or combinatorial cuts to each master program using a local branching mechanism. Saharidis, Minoux, and Ierapetritou (2010) augmented Benders cuts for scheduling problems by generating covering cut bundles, which jointly involve many complicating variables and thus provide the beneficial effect inherent with using high-density cuts. Another work in this vein (by Saharidis and Ierapetritou 2010) generated an additional Benders cut by solving a maximum feasible subsystem whenever a feasibility cut is generated, where a significant improvement in the convergence behavior was demonstrated for certain scheduling problems that involve more feasibility than optimality cuts.

As an alternative, Sherali and Lunday (2013) proposed a perturbation scheme for the subproblem that automatically generates maximal-nondominated Benders cuts via a single subproblem optimization step, which is also convenient for implementation. Following this strategy for Problem IPM$_h$, $\forall\, h \in H$, we perturb the RHS of Constraints (60) and (61) as specified below

$$\sum_{p\in\Pi_j} \pi_{ph} \le \sum_{k\in K} \sum_{a\in\delta_j^-\cap A^k} \tilde{\gamma}_{jkh}(\bar{x}_a + \epsilon\hat{x}_a), \quad \forall\, j \in L, \tag{69}$$

$$0 \le \pi_{ph} \le (1+\epsilon)\tilde{\mu}_{ph}, \quad \forall\, p \in \Pi, \tag{70}$$

where $\hat{x}$ is a predefined positive weight vector (we use $\hat{x}_a = 1$, $\forall\, a \in A$), and where ϵ is a perturbation coefficient (we use $\epsilon = 10^{-6}$). Note that the dual solution generated for the perturbed subproblem is also feasible to the original dual subproblem. Thus the resulting Benders cut remains valid. However, the dual objective function of the perturbed subproblem inherits a corresponding perturbation term that guides the selection of a dual solution among (near-) alternative optimal solutions to derive maximal nondominated Benders cuts. Likewise, for Problem CP^f, $\forall\, f \in F$, we rewrite Constraints (64) and (65) with appropriate perturbations as

$$\sum_{p\in P^f} e_{jp}^f z_p^f + q_j^f = \sum_{k\in K^f} \sum_{a\in\delta_j^-\cap A^k} (\bar{x}_a + \epsilon\hat{x}_a), \quad \forall\, j \in L, \tag{71}$$

$$\sum_{p\in P^f} s_{\sigma p}^f z_p^f + q_\sigma^f = \sum_{a\in\mathcal{A}_\sigma^f} (\bar{x}_a + \epsilon\hat{x}_a), \quad \forall\, \sigma \in S^f, \tag{72}$$

and we proceed with the solution approach described above, wherein the resulting dual solutions are used to derive the Benders cuts (62) and (67) in the same form as before (i.e., without the perturbation terms).

In addition, we adopted a tabu-type strategy to help alleviate an observed stalling phenomenon for the RMP by suppressing repetitive Benders subproblems. Because of the inherent symmetry in the model, distinct $\bar{x}$-solutions could lead to identical Benders subproblems by providing the same RHS values. Specifically, since Constraints (59)–(61) in Model IPM are essentially determined by the fleet assignment decisions $\bar{y}_{jk} \equiv \sum_{a\in\delta_j^-\cap A^k} \bar{x}_a$, different $\bar{x}$-values could result in the same fleet assignment, thus creating the same subproblem. Therefore, we keep track of the distinct fleeting decisions using a tabu list, and invoke the subproblem only when the $\bar{x}$-values indicate a new fleet assignment pattern. Likewise, Subproblem CP (given by (63)–(66)) is invoked only when any flights are assigned to a different aircraft family, or if the involved short connections are changed.

5. Computational Results

In this section, we present numerical results using a series of test instances based on historical data obtained from *United Airlines*, a U.S. legacy airline carrier. For each of the test scenarios, a two-phase procedure was adopted, i.e., we solve each of the test scenarios using the sequential approach and the proposed integrated FRC model, where in the former, the partially integrated Model FAAR-RLT is solved before the conventional crew pairing model. This solution framework was implemented using C++ along with the off-the-shelf MIP solver CPLEX 12.5 with default settings (except as noted). All tests were performed on a desktop computer with dual quad-core Intel Xeon 2.4 GHz CPUs with 24 GB RAM and running a 64-bit Windows system.

5.1. Data Description

The instances used for our experiments were derived from the United Airlines North American flight schedule based on its most frequently served airports. Because the network has dense flights between such hubs and major stations that attract many travelers, there are a large number of connection opportunities for aircraft as well as crew. Note that arcs of types A_5 and A_6 (arrivals that connect to third-day departures) were omitted when generating connections, since they are unfavorable from the perspective of an efficient schedule and therefore rarely occur in practice.

To cover the flight network, we principally considered the mid- and large-sized jets, e.g., Airbus 320 series (including fleets of Airbus 319 and 320), and Boeing 757 (including fleets of Boeing 757–200 and 757–300) and a fleet of Boeing 777–200 series used for the domestic service. Given the hourly fuel consumption rate for a fleet type, the flight operational cost was estimated using the flight duration, which was further incremented by a fixed percentage (5%) to account for any unplanned detours.

Fleets that are customized for domestic flights usually have three cabin classes, i.e., business class, economy plus class that provides extra leg room, and regular economy class. For the sake of simplicity, we assumed that travel demands for each origin-destination (O-D) pair are categorized, and therefore priced according to these actual cabin classes, although in practice there are a variety of fare classes for each cabin. Moreover, for each fleet type, the number of seats in each class was taken as fixed and independent (i.e., not nested). In the planning process, as time approaches the departure date, an increasing load factor (e.g., 80% and above) is used to discount the fleet capacity for the purpose of reserving seats for close-in bookings. Whenever the demand for a particular flight-cabin on any flight leg exceeds the corresponding effective capacity on the assigned aircraft, we assume that a certain proportion (20%) of the spilled passengers are lost; this was

Table 1 Description of Test Instances

	Flights	Stations	Total connections			Short conn. for A320	Aircraft count			Total itineraries
			A320	B757	B777		A320	B757	B777	
HS1	128	17	4,728	4,586	4,313	171	17	17	1	1,642
HS2	154	10	8,679	8,395	8,133	142	20	20	1	1,315
HS3	246	19	18,673	18,128	17,552	324	36	36	2	3,516
HS4	354	25	23,742	23,066	22,209	585	57	57	4	6,124
HS5	440	24	38,587	37,508	36,127	782	60	60	4	7,976
K1	205	11	10,596	10,327	10,094	208	33	33	2	2,319
K2	282	15	15,488	15,118	14,736	304	42	42	3	4,275
L1	522	23	44,435	43,629	41,645	274	75	75	6	9,075
L2	676	33	62,152	60,614	58,437	1,230	92	92	8	14,014

accordingly used to account for the opportunity cost factor in (1). As noted earlier, these assumptions do not reflect all of the dynamics for the passenger spill and recapture process among fare classes and itineraries, nor do they consider day-to-day demand variations; the resulting revenue only accounts for part of the realized revenue achieved by an airline. However, this affords a simple but reasonable approximation for early-stage planning decisions; more sophisticated models can be similarly integrated for real-life practice.

Moreover, each fleet type has its own maintenance requirements between two consecutive A-checks, which govern the specified maximum flying hours ($t_{\max}$), the maximum number of days ($d_{\max}$), and the maximum number of take-offs ($to_{\max}$). Our data sets and implementation focused on monitoring and enforcing the maximum flying days requirement, where the restriction on the consecutive flying minutes is additionally automatically guaranteed by assuming that $1{,}440 \times d_{\max} \leq t_{\max}$.

As noted in §1, crew are classified according to the aircraft family on which they are eligible to serve. Crew costs are measured in hours, and determined by a minimum guaranteed pay (assumed to be based on four hours a day), the actual flying time, and a proportion of the total TAFB (assumed to be 80%). Furthermore, to compatibly simplify crew costs with other expenditures, we used a fixed rate to estimate the total hourly salary of pilots and flight attendants serving a pairing. Also, we designated eight particular stations as crew bases, so that crew pairings must start and end at one of these stations. In addition, we assume that crews can serve at most four flights in a duty, and at most two to four days in a pairing. Moreover, the separation time between duties must be at least 10 hours. While these assumptions are not as specific as those imposed in practice, a richer list of crew rules can likewise be adopted, which we recommend for future research.

Nine instances, as presented in Table 1, were generated for testing the proposed solution methodology. These instances consider two types of networks, the hub-and-spoke and the point-to-point, that include major airports in the continental United States. Here, we only consider the U.S. major airports that involve several daily flights because (i) they address a significant portion of the daily travel demand, and (ii) for other smaller spoke airports, the flights are usually less frequent and well scheduled, and thus can be appropriately preprocessed. The number of connections displayed in Table 1 reflects the number of binary variables for each designated fleet type. Also, for the sake of illustration, we present the number of available short connections for Airbus aircraft. Although a typical full-scale daily domestic network has more flight legs than our largest data instance, our purpose here is to focus on suitable dense subnetworks that represent principal connection opportunities. Hence, these instances are significant and challenging.

5.2. Results for the Partially Integrated Approach

We first present computational results for Phase I of the proposed algorithm, which involves solving the FAAR problem and then sequentially solving the CP problem separately for each aircraft family based on the aircraft assignments obtained from the solution to Model FAAR-RLT. Table 2 reports the results obtained for Model FAAR-RLT, and Table 3 provides the results for the subsequent solution of Model CP.

From a numerical perspective, the results demonstrate the efficacy of the partially integrated Model FAAR-RLT, wherein all data instances were effectively solved within reasonable computational times that are proportional to the problem size. The solution times required for the largest data instances were about 5–10 hours. When solving the master program as an MIP, the Benders subproblem routine is used to find an integer solution whose objective value within the restricted master program is greater than the current incumbent value by more than a pre-specified threshold of 1%; otherwise, it is not invoked as it has limited potential for improvement. The penultimate column in Table 2 specifies the number of Benders subproblems solved. The actual number of the aforementioned integer solutions detected is given in parentheses.

Table 2 **Fleet Assignment and Aircraft Routing Results**

FAAR	Flights covered by A320	B757	B777	Aircraft used	LB on required aircraft (§3.1.1)	Comp. time (min)	Benders SP (int. sol. found)	Opt. gap
HS1	66	60	2	35	35	0.79	1(3)	Opt
HS2	102	52	—	40	39	4.89	1(2)	Opt
HS3	133	109	4	74	69	45.77	2(3)	0.74%
HS4	193	147	14	118	113	42.61	1(5)	0.06%
HS5	256	172	12	124	120	124.64	1(5)	Opt
K1	134	71	—	66	65	7.12	1(4)	0.12%
K2	180	99	3	86	83	21.64	1(5)	0.01%
L1	316	192	14	156	148	376.19	1(5)	0.03%
L2	388	265	23	192	190	594.80	1(6)	0.01%

Table 3 **Crew Pairing Results**

CP	Pairings generated A320	B757	B777	Short conn.	Short conn. (%)	Pairing cost (hr)	Cost per flight (hr)	Comp. time (min)	Obtained at/via:
HS1	522	308	3	19	14.8	503.14	3.93	0.02	Root
HS2	2,495	248	—	31	20.1	579.65	3.76	0.25	Root
HS3	3,401	933	7	16	6.5	1,056.28	4.29	0.47	Root
HS4	3,721	977	177	47	13.3	1,772.98	5.01	1.77	IP
HS5	6,271	1,640	74	63	14.3	1,960.33	4.46	5.54	IP
K1	3,174	884	—	33	16.1	814.43	3.97	1.16	IP
K2	4,084	829	4	48	17.0	1,067.50	3.79	1.48	IP
L1	8,928	1,882	113	73	14.0	2,482.02	4.75	13.43	Root
L2	13,445	3,666	168	111	16.4	3,542.63	5.24	34.47	IP

We also compared the lower bound on the minimum number of required aircraft for each family, as computed in §3.1.1, with the actual number given by the solution. Here, note that the lower bounds were all generated by solving the LP relaxation of the problem to minimize the total number of required aircraft subject to Constraints (42)–(53). Apparently, the best solutions use more aircraft than needed to run the network efficiently and to maintain a cost-effective schedule. Furthermore, among the three aircraft families, because the Airbuses have the lowest hourly operating costs and the shortest turn-times, they have the highest utilization rate over all test instances, and cover a large proportion of flights in the network. On the other hand, for the given test data, the bigger aircraft, e.g., the Boeing 757 series, were used to a lesser extent, and the Boeing 777 aircraft were seldom deployed.

Given the aircraft assignments to the flight legs as determined by the solution to Model FAAR-RLT, the CP model was then solved using the branch-and-price heuristic **BPH** for each family. Here, to curtail the solution effort, the solution to CP was terminated on finding the first integer-feasible solution. Moreover, whenever an integer solution was not obtained after generating a sufficiently large number of pairings (set as 15 times the number of flights in the subnetwork), we switched to solving an integer program (IP) using the generated pairings. This strategy was well justified in our computational experience since the resultant gap between the final objective value and the LP relaxation value turned out to be sufficiently small ($<0.5\%$). The results presented in Table 3 show that the methodology solved the test instances relatively fast. In general, difficult cases adopted the restricted IP strategy in the solution process while smaller problems tended to be solved at the root node depending on the structure of the problem. This influences the number of feasible crew pairings generated at the root node by the constrained shortest path algorithm, as displayed in the table. Furthermore, the results show that up to 20% of connections are short.

To assess the quality of the crew pairing results, the CP cost (converted to hours) was also benchmarked against the total number of flights for the given instance. Not surprisingly, the network involving only the key hubs yielded a relatively smaller cost per flight duration ratio since (i) at each station, there are many connection opportunities that involve short sit-times, and (ii) these key hubs are mostly crew bases, which saves on overnight costs. On the other hand, for the larger hub-and-spoke network, the CP cost per flight hour turned out to be relatively higher, but could be reduced by recruiting crew from an extended list of stations.

Table 4 Results for the Integrated Approach

	Flights covered by						
FRC	A320	B757	B777	Aircraft used	Comp. time (min)	Benders SP (int. sol. found)	Improvement over phase I (%)
HS1	75	51	2	35	0.98	1(1)	9.0
HS2	94	60	—	40	6.79	1(4)	11.3
HS3	132	108	6	74	24.56	1(1)	5.4
HS4	190	150	14	118	37.45	1(8)	16.1
HS5	247	182	11	124	140.81	1(7)	1.4
K1	139	66	—	66	8.49	1(2)	6.2
K2	179	103	—	84	43.36	1(5)	13.7
L1	299	206	17	156	250.68	1(5)	5.9
L2	383	270	23	192	872.35	1(1)	6.3

Table 5 Results for the CP After FRC

	Pairings generated							
Final CP as an IP	A320	B757	B777	Short conn.	Pairing cost per flight (hr)	Cost change w.r.t. phase I (%)	Comp. time (min)	Obtained at/via:
HS1	674	155	3	19	3.86	−1.9	0.03	Root
HS2	1,772	439	—	19	3.63	−3.5	0.08	Root
HS3	3,514	885	14	18	4.26	−0.8	0.45	Root
HS4	3,395	954	236	46	4.98	−0.5	1.05	Root
HS5	5,610	1,644	33	73	4.44	−0.2	3.63	BPH
K1	3,961	599	—	38	3.88	−2.4	0.68	IP
K2	4,047	1,120	—	44	3.62	−4.3	12.50	BPH + IP
L1	7,057	1,788	99	78	4.38	−7.9	8.73	IP
L2	10,010	3,209	189	118	4.46	−14.9	28.00	BPH + IP

5.3. Results for the Integrated Fleeting, Routing, and Crew Pairing Approach

We next present in Table 4 numerical results for Phase II of the proposed algorithm, which involves solving the integrated Model FRC. Because this solution process repeatedly solves the passenger-mix and CP subproblems as opposed to the sequential approach of Phase I, it consumes a relatively greater computational effort, ranging up to 15 hours. Note that this solution process benefits from Phase I in that an initial solution is provided to the integrated problem, which curtails early calls to Benders subproblems. The penultimate column of Table 4 displays the number of Benders subproblems solved. The actual number of the aforementioned integer solutions detected is given in parentheses. The extra effort expended by the integrated model is worthwhile since a significant improvement is achieved in the objective function value over that from the partially integrated Phase I solution, where this results from a combination of reduced crew pairing costs and increased revenues from passengers. In addition, as presented in Table 5, the CP problem is finally solved as an IP using the same methodology as in Phase I, the results of which show that the larger jets, i.e., the Boeing 757 series and 777 series, were assigned to more flight legs so as to facilitate improved crew pairing decisions and thus lower the overall costs for a few cases. Although the aircraft operational costs were subsequently higher, the look-ahead mechanism afforded by the integrated approach compensated for this increase by lowering crew costs, thus resulting in an overall cost reduction. It can be reckoned that the improvement over the conventional sequential approach (which further separates the solution of the fleet assignment and the aircraft routing problems) would only be greater.

5.4. Analysis of Acceleration Strategies

To demonstrate the benefits of the adopted acceleration strategies, we conducted some experiments by turning off each of these strategies in turn. Below we focus on two particular strategies that made a significant difference, i.e., generating nondominated cuts per Sherali and Lunday (2013) and implementing the initial cut generation strategy of McDaniel and Devine (1977). Note that we still retain and advocate the other strategies since they helped somewhat. Note also that these strategies might be relevant for other data sets as evident from the papers that proposed them. Moreover, the corresponding computational results reported in Table 6 indicate that the performance of the proposed algorithm can be severely impaired without such strategies. Specifically, when turning off the nondominated cut generation strategy, the increase in terms of computational times ranged up to 13.7%, except for the largest instance that was not solved within the prescribed time limit (i.e., 10 hours) without this strategy. Note that the relatively small increment

Table 6 **Computational Effort Without Some Key Acceleration Strategies**

	Turning off nondominated cuts		Turning off initial cut generation	
FRC	Comp. time (min)	Increase w.r.t. the proposed alg. (%)	Comp. time (min)	Increase w.r.t. the proposed alg. (%)
HS1	1.00	2.4	6.29	546
HS2	7.17	5.6	9.78	44
HS3	25.73	4.8	—	—
HS4	41.78	11.6	—	—
HS5	147.82	5.0	—	—
K1	9.41	10.9	28.17	232
K2	46.49	7.2	98.08	126
L1	284.97	13.7	—	—
L2	—	—	—	—

in computational time is due to the fact that the crew pairing subproblem was sparingly invoked (see Table 4). Moreover, when skipping the initial cut generation approach of McDaniel and Devine (1977), only four small models were solved within the 10-hour time limit, where the computational times for Model FRC increased by an amount ranging from 44% to a factor of 5.46.

6. Summary and Conclusions

In this paper, we have presented an integrated model along with associated solution methodologies for the airline scheduling process. The proposed model incorporates itinerary-based fleet assignment, aircraft routing, and crew pairing within a single framework. The model therefore extends the node-arc formulation for the aircraft routing problem proposed in Haouari, Shao, and Sherali (2013) to incorporate fleet assignment decisions, while retaining a compact, polynomial-sized linear MIP. Furthermore, this model includes additional features of itinerary-based demands along with crew pairing decisions, where the latter therefore simultaneously generates minimum-cost rotations for crew groups according to their eligible aircraft families.

To effectively address this integrated Model FRC, we adopted a Benders decomposition solution framework wherein the fleeting and routing decisions were treated within the master program, and the passenger-mix and crew pairing decisions were handled via separate subproblems. In particular, we further embedded a column generation algorithm within a Lagrangian relaxation scheme, where a deflected subgradient algorithm (Subramanian and Sherali 2008) was applied to solve the restricted master program, and where eligible pairings were derived using a constrained shortest path algorithm. Moreover, we generated maximal nondominated Benders cuts per Sherali and Lunday (2013) via the passenger-mix and crew pairing subproblems, and we adopted several other acceleration strategies such as the Benders scheme proposed by McDaniel and Devine (1977) and Geoffrion and Graves (1974) to generate an initial set of cuts and to effectively manipulate the master program, respectively.

To demonstrate the benefits of this integrated modeling approach, we proposed a two-phase solution framework, where in Phase I we solved Model FAAR-RLT that represents a partially integrated itinerary-based FAAR problem, followed by a sequential solution of a CP problem for each sub–network based on the resulting fleet assignment decisions. In Phase II of this framework, we solved the fully integrated Model FRC as described above, using a set of initial cuts as generated during Phase I. Several test instances were derived using historical data obtained from United Airlines, a legacy U.S. carrier. The computational results obtained demonstrated the value of adopting an integrated viewpoint in Phase II as opposed to the (partially) sequential approach of Phase I. On average over our set of nine test instances, this achieved an improved profit of about 8.4%, which accrued mainly from a judicious, albeit more expressive, utilization of the available fleet in a manner that significantly decreased crew pairing costs, thus reducing overall costs. Note that the improvement mentioned here is also constrained by the accuracy of early-stage demand forecasts and the realism of the adopted passenger-mix model. A more sophisticated passenger-mix model as discussed in §3 could serve to better address real-life customer choice dynamics.

The flexibility of the proposed model permits incorporating other aircraft routing restrictions without adding much expense in terms of the model size. In addition, the crew pairing rules can be readily modified to accommodate different realistic requirements. To improve the accuracy of profit predictions, it is also possible to accommodate further enhanced features that consider an itinerary-based spill and recapture component, and to tie in the model with more sophisticated revenue management strategies to better address seat inventory control, as well as to provide insights into pricing and overbooking decisions. Moreover, other airline operational decision-making stages can be further integrated in the developed framework, such as the *demand-driven dispatch* strategy (Shebalov 2009) that examines provisional plane swaps according to updated demand information. Furthermore, the proposed framework can be used to study other higher-level decisions, for example, the potential benefits of incorporating certain optional flight legs. Finally, another possible direction for future research is to enhance the model's solvability from the computational perspective by using parallel computing techniques, particularly since Benders subproblems are solved independently from each other.

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